$$\mathbf{v} = (I_N \otimes W_v^T)\mathbf{x} \quad (N = 3,\ D_v = D)$$

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \left(\begin{array}{c|c|c} W_v^T & 0 & 0 \\ \hline 0 & W_v^T & 0 \\ \hline 0 & 0 & W_v^T \end{array}\right) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$I_3 \otimes W_v^T$$

$$\mathbf{y} = (A(X) \otimes I_D)\mathbf{v}$$

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \left(\begin{array}{c|c|c} a_{11}(X)I & a_{12}(X)I & a_{13}(X)I \\ \hline a_{21}(X)I & a_{22}(X)I & a_{23}(X)I \\ \hline a_{31}(X)I & a_{32}(X)I & a_{33}(X)I \end{array}\right) \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$A(X) \otimes I_D$$

$$M(X) = [a_{nm}(X)W_v^T]_{n,m} = A(X) \otimes W_v^T$$